Machine Learning with Graphs: Projects

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A semester-long class project will be an important part of this course. The goal of the project is to allow students to have a hands-on (coding) experience applying the topics covered during the semester.

Projects should be done in groups of up to three students. Each student should list their specific contributions to the project reports.

The projects will be evaluated based on the creativity of the solution proposed, the relevance of the topic to the course, and the overall quality of the final report. Implementations should be made publicly available. Notice that the project will make up 40-50% of the final grade for this course. Next, the deliverables for the project are summarized next.

Project Proposal (02/17)

A two-page description of the problem to be addressed in the project and the relevant related work. A typical project will be based on a dataset and/or a published paper and will address one of the following:

1. **Emerging GraphML application:** Select an application (problem and dataset) where GraphML has not been extensively applied, apply an appropriate model, and analyze the results.

2. **Evaluating a recent GraphML model:** Select a recent GraphML model, identify an appropriate dataset to which the model has not been applied yet, and analyze the results.

3. **Addressing a GraphML problem in a new setting:** Select a GraphML problem and a setting not extensively studied, propose a modification of an existing GraphML model that attempts to address the new setting, and analyze the results.

4. **Novel synergies between Large Language Models and graphs:** Select an ML problem to which a combination of GraphML and LLMs has not been extensively applied to, propose an appropriate solution, and analyze the results.

The proposal should include a motivation, the problem to be addressed, and a timeline indicating when the major steps of the project are expected to be finished.

Progress Report Meeting (3/24)

Each group will schedule a meeting with the instructor to discuss the progress towards the project. The meeting should cover feedback received about the proposal, challenges faced, preliminary results, and updated project timelines and goals.

Final Report (05/05)

The final report should summarize the entire project, including references, main challenges, results, and discussion. The length of final reports is limited to eight pages (excluding references) using the Neurips format.[1] Reports are expected to be original, clear, grammatically correct, well-organized, and technically sound. The contributions made by each student in the group should be summarized in the final report.

[1] `https://nips.cc/Conferences/2020/PaperInformation/StyleFiles`